SENTHILKUMAR M

AI / Machine Learning Engineer (LLMs & RAG)

8667254699 kumarsenthil41073@gmail.com Salem, Tamil Nadu 636203, India

SUMMARY

As an AI Engineer specializing in Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG), I have a background in developing end-to-end AI solutions. My expertise lies in building RAG pipelines, utilizing embedding te semantic retrieval, and deploying systems using FastAPI. I am adept in Python and various libraries, ensuring efficier of generative AI technologies

PROJECTS

RAG-Based Question Answering System

A system that answers questions using retrieval-augmented generation methods

- Designed and implemented an end-to-end Retrieval-Augmented Generation (RAG) pipeline
- Performed document chunking and generated embeddings for semantic retrieval
- Integrated vector database (FAISS) to retrieve relevant context for LLM responses
- Improved answer accuracy and reduced hallucination compared to standalone LLM

LLM-Powered PDF Chatbot

A chatbot that allows users to interact with PDF documents through natural language

- Built an AI chatbot enabling natural-language queries over PDF documents
- Implemented semantic search using embeddings and vector similarity matching
- Engineered prompts to generate context-aware and grounded responses
- Deployed the application as a REST API using FastAPI

Embedding-Based Semantic Search Engine

A search engine that uses embeddings for more relevant search results

- Built a semantic search system using dense embeddings and vector similarity
- Enabled meaning-based retrieval beyond keyword search
- Evaluated retrieval relevance and optimized chunk size for performance

Prompt Engineering & Evaluation Project

A project focused on prompt engineering to enhance large language model outputs

- Designed and tested prompt strategies to improve factual accuracy
- Reduced hallucination using context grounding and structured prompts
- Compared LLM outputs with and without retrieval support

SKILLS

Python (NumPy, Pandas, OOP) | SQL (basic) | RAG | Prompt Engineering | ChromaDB | Document Chunking | Document Chunking | LangChain | Hugging Face | NLP

CERTIFICATIONS

HackIndia 2025 – AI & Web3 Hackathon | HackIndia

Retrieval-Augmented Generation witl (Lab) – IBM SkillsBuild

Getting Started with Artificial Intelligence – IBM SkillsBuild

Hugging Face – Transformers Course

Journey to Cloud: Envisioning Your Solution – IBM SkillsBuild

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